MILITARY SPECIFICATION

BELT, AIRCRAFT SAFETY, HBU-12/A

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of aircraft safety lap belt for the aircraft ejection seat systems described in 6.1. The belt is designated HBU-12/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-6021	Castings, Classification and Inspection of
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-A-83469	Adjuster, Lap Belt, 1-3/4 Inch Webbing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1680

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STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Frost Engineering Development Corporation (FSCM 29645)

549015	Gold Key, Buckle Hardware
549100	HBU-12/A Lap Belt Assembly
549110	Webbing Assembly, HBU-12/A Lap Belt Assembly
549120	Manual Release Assembly, Buckle Hardware
549130	Automatic Disconnect Link Assembly, Buckle Hardware
549140	Buckle Hardware, HBU-12/A Lap Belt Assembly

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test Methods for
ASTM E192	Reference Radiographs for Investment Steel Castings for Aerospace Applications
ASTM D3951	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publication are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.5 and 6.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials which are used in the fabrication and assembly of the lap belts and buckle hardware shall comply with the requirements of this specification. Materials which are not specifically designated, or which are offered as equivalent to specified manufacturer's part numbers shall require approval by the procuring activity.

3.3.1 Annealing and heat treating. The annealing and heat treating of stainless steel parts shall be accomplished in a vacuum furnace or an inert atmosphere.

3.4 Design and construction. The aircraft safety lap belt assemblies shall be in accordance with Drawing 549100, the subsidiary drawings listed below and their detail drawings, and the requirements specified herein. If the requirements of the drawings conflict with the requirements of this specification, the requirements of this specification shall govern.

DRAWINGS	COMPONENTS	QUANTITY
549110	Webbing Assembly, HBU-12/A Lap Belt Assembly (one left hand (L.H.) and one right hand (R.H.))	1 Set
549140	Buckle Hardware, HBU-12/A Lap Belt Assembly	1 Set

3.4.1 Buckle hardware. The buckle hardware (Drawing 549140) shall consist of a manual release assembly, an automatic disconnect link assembly and, when applicable, a gold key.

3.4.1.1 Manual release assembly. The manual release assembly shall conform to Drawing 549120 and shall have provision for connection of the L.H. webbing assembly. The design shall be such that the release handle cannot be closed without having the gold key locked to the assembly. However, it shall also be possible for organizational-level personnel to modify the assembly easily so that the handle can be closed without the gold key, for belts used in ejection seat systems that do not need it. In either case, it shall be impossible to open the release handle without squeezing its grips together by applying forces to them simultaneously in opposite directions.

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3.4.1.2 Automatic disconnect link assembly. The automatic disconnect link assembly shall conform to Drawing 549130 and shall include provisions for connection of the R.H. webbing assembly and a hot-gas hose. The link assembly shall incorporate a disconnectable tongue for engaging it with the manual release assembly after the tongue has been inserted through restraint strap end loops and the gold key (when one is used). The L.H. end of the tongue shall be shaped to separate without hangup from the gold key and the restraint straps when the manual release is operated, but to hold the gold key locked to the manual release when the link assembly and restraint straps separate from the R.H. end of the tongue in response to hot-gas actuation of the automatic disconnect mechanism.

3.4.1.3 Gold key. The gold key shall conform to Drawing 549015 and shall have provisions for operating the mechanism in the manual release assembly which prevents handle closure unless the gold key is locked to the assembly. The design shall provide for attachment of a parachute-opening lanyard to the gold key and for inserting through it the L.H. tip of the link assembly tongue, but not its body.

3.4.2 Webbing assemblies. The webbing assemblies (Drawing 549110) shall have provisions for attaching their outboard ends to the ejection seats and for adjustment of their length. The R.H. assembly shall incorporate means for holding the hot-gas hose closely against it (see 3.4.1.2).

3.4.2.1 Adjusters. The adjusters incorporated in these webbing assemblies shall be of a type which has been qualified to MIL-A-83469 requirements while installed on webbing of the type used in these assemblies. The adjusters shall also be capable of fulfilling the adjuster performance requirements of this specification.

3.5 Performance characteristics

3.5.1 No-load manual opening force. With tensile load of not more than 10 pounds applied to the coupled release and link assemblies and with the latch handle fully depressed, the actuating force applied to the hook handle to disconnect the manual release assembly from the automatic disconnect assembly shall be not less than 2 nor more than 12 pounds (see 4.7.3).

3.5.2 Static-load manual opening force. With static tensile load of 300 pounds applied to the coupled release and link assemblies, and with the latch handle fully depressed, the force applied to the hook handle to disconnect the manual release assembly from the automatic disconnect assembly shall be not less than 2 nor more than 30 pounds (see 4.7.4).

3.5.3 Static-load automatic disconnection pressure. With static tensile load of 1000 pounds applied to the coupled release and link assemblies, the pressure required to effect automatic disconnection shall be at least 75 psig, but shall not exceed 750 psig (see 4.7.5).

3.5.4 Automatic disconnect leakage. No leakage shall occur when an actuation pressure of 2000 psig is applied to the disconnected link assembly and sustained for 30 seconds (see 4.7.5.1).

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3.5.5 Manual release cycling. The buckle hardware shall be capable of withstanding 2500 cycles of engaging and disengaging the release and link assemblies with 50-pound tensile load applied to them during the disengaging portion of each cycle. This life-cycling test shall not cause excessive wear and the no-load opening force shall be in accordance with the requirements of 3.5.1 when checked after every 500 cycles (see 4.7.6).

3.5.6 Vertical drop. The coupled release and link assemblies shall be capable of withstanding six 8-foot vertical drops onto concrete without structural failure or damage sufficient to prevent demonstration thereafter that no-load manual release and static-load automatic disconnect operation are in accordance with the requirements of 3.5.1 and 3.5.3 (see 4.7.7).

3.5.7 Vibration. The coupled release and link assemblies shall be capable of withstanding the effects of vibration in the 10 to 55 Hz range. This shall not cause disconnection nor shall any of their parts work loose, and operation of the manual release and automatic disconnect assemblies thereafter shall be in accordance with the requirements of 3.5.1 and 3.5.3 (see 4.7.8).

3.5.8 Shock/impact. The coupled release and link assemblies shall be capable of withstanding the effects of 400g ±40g shock/impact. This shall not cause disconnection nor shall any of their parts work loose, and operation of the manual release and automatic disconnect assemblies thereafter shall be in accordance with the requirements of 3.5.1 and 3.5.3 (see 4.7.9).

3.5.9 Buckle hardware ultimate tensile strength. The coupled release and link assemblies shall sustain a static tensile load of 7000 pounds for 30 seconds without disconnection. Permanent deformation of components is permissible, but there shall be no evidence of structural failure, and operation of the manual release assembly after load removal shall be in accordance with 3.5.1 (see 4.7.10).

3.5.10 Automatic disconnect ultimate pressure retention strength. The pressure-carrying portion of the automatic disconnect link assembly shall sustain internal hydrostatic pressure of 7500 psig for 5 seconds without structural failure or leakage of more than 0.5 cc of fluid (see 4.7.11).

3.5.11 Gold key proof load. With the gold key installed on a link assembly tongue and the latter engaged with the release assembly, the gold key shall sustain for 5 seconds a removal force of 75 pounds applied to its lanyard attaching provision in the direction of the tongue. There shall be no effect on the key or the buckle hardware to which the key is attached (see 4.7.12).

3.5.12 Gold key ultimate strength. With the gold key installed on the link tongue coupled to the release assembly and load applied as in 3.5.11, the gold key shall sustain for 30 seconds a removal force of 450 pounds. There shall be no effect on the buckle hardware components nor structural failure of the gold key, but permanent deformation of it is permissible provided that it does not pull off the tongue (see 4.7.13).

3.5.13 Webbing length adjustment force. The tensile force applied to the free end of a webbing assembly to produce 25-pound tensile load in buckle hardware connected to it shall not exceed 50 pounds. The force applied to the free end of the fabric tab of the adjuster assembly to release this 25-pound tensile load in the buckle hardware shall not exceed 15 pounds (see 4.7.14).

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3.5.14 Webbing assembly proof load. With a minimum of 2 inches between the adjuster and the doubled-over free end of the webbing emerging from it, each webbing assembly shall sustain a static tensile load of 1000 pounds for 10 seconds. This proof load shall have no effect on the anchor adjuster, webbing, or its sewing, nor shall slippage of the webbing through the adjuster exceed 1/2 inch (see 4.7.15).

3.5.15 Webbing creep. Creep of webbing through either adjuster of a belt assembly shall not exceed 1/2 inch when the complete assembly is subjected to 2500 cycles of tensile loading that is cycled from 0 to 226 pounds at the rate of 60 ±10 cycles per minute (see 4.7.16).

3.5.16 Combined proof loads. With a negative-g strap and two shoulder harness straps properly engaged with the coupled release and link assemblies, the lap belt shall sustain for 30 seconds simultaneous static tensile loads of 2880 pounds applied via its webbing assemblies and 3600 pounds applied via the opposed negative-g and shoulder harness straps in the same plane as the buckle hardware, but at right angles to the belt load. These combined proof loads shall not cause the webbing to slip more than 1/2 inch through either adjuster nor shall structural failure or permanent deformation occur in any component of the belt assembly. Operation of the manual release and automatic disconnect assemblies after this proof loading shall be in accordance with 3.5.2 and 3.5.3 respectively (see 4.7.17).

3.5.16.1 Combined proof loads without negative-g strap. The foregoing requirements are also applicable to the loading condition in which the shoulder-harness loads are not counterbalanced by a negative-g strap. The lap belt shall be capable of sustaining for 30 seconds the 2880-pound load in the webbing assemblies and an 1800-pound load in each shoulder harness strap, applied simultaneously to the coupled release and link assemblies and in essentially the same directions as when the belt and shoulder harness straps are installed on an ejection seat and are restraining its occupant against sustained X-axis deceleration (see 4.7.17.1).

3.5.17 40-g dynamic load. With or without a negative g strap, the lap belt assembly shall be capable of withstanding dynamic loads such as those imposed on it in a crash or the aerodynamic deceleration phase of emergency escape. The loads shall be typical of those produced in such circumstances by 40-g X-axis deceleration of a 95th percentile size user (as defined by the 1967 USAF Flying Population Survey), restrained by the belt, in conjunction with two shoulder harness straps. All equipment shall be installed correctly on a seat of adequate strength, and properly interconnected and tensioned. There shall not be any structural failures of the lap belt assembly nor webbing slippage that exceeds 1/2 inch per adjuster. Permanent deformation and other such non-castastrophic reactions of lap belt components that would not cause injury to the restrained occupant are permissible. The lap belt buckle shall be operable in accordance with the requirements of 3.5.1 to permit manual release of the restraint system immediately after imposition of the dynamic loads has ceased (see 4.7.20).

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3.5.18 Environmental requirements. The buckle hardware and lap belts shall be capable of demonstrating compliance with the performance requirements described in 3.5.1 through 3.5.17 after exposure to any of the following conditions.

3.5.18.1 Temperature. Maintenance of adjusted length without significant slippage under high tensil proof load, manual release, automatic disconnection, and webbing length adjustment of the lap belt assembly shall be performable over the entire temperature range from -65°F to +200°F (see 4.7.18.1 and 4.7.18.2).

3.5.18.2 Humidity, salt fog, and dust. Manual release, automatic disconnection, and webbing length adjustment of the lap belt assembly shall be performable after it has been exposed to humidity, salt fog, and dust environments (see 4.7.19, 4.7.19.1, and 4.7.19.2).

3.6 Interchangeability. There shall be complete interchangeability with respect to mating of the release and link assemblies and the gold key of the buckle hardware. All other components having the same part numbers shall also be interchangeable with each other with respect to form, fit, and function. Changes in any part shall be governed by the drawing number requirements of DOD-STD-100, but none shall be made without prior approval of the responsible Government engineering activity obtained via the contracting activity.

3.7 Identification marking. Marking for identification shall be in accordance with MIL-STD-130, and drawings referenced herein.

3.7.1 Nameplate. Identification of the buckle hardware shall be by nameplate applied to the outer surface of the release assembly base plate. It shall be permanently marked with the following information as a minimum.

a. Belt, Aircraft, Safety HBU-12/A.

b. Part Number (including the applicable revision letter) of the release assembly.

c. Month and year of manufacture (both may be shown in abbreviated numeric form if desired, such as 6/80 instead of June 1980).

d. National Stock Number (when assigned).

e. Contract or purchase order number.

f. Manufacturer's identification Federal Supply Code for Manufacturers (FSCM) number as a minimum).

g. Serial and specification numbers when specified by the contracting activity.

3.7.2 Part numbering. The automatic disconnect link assembly and its tongue, the adjusters, anchor fittings, and removable webbing crossbars shall be permanently marked with their respective part numbers (including revision letter). The link assembly and the adjusters shall also be serial numbered or marked with the month and year of their manufacture (abbreviated as in 3.7.1, if desired), and the manufacturer's identification as a minimum.

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3.7.3 Nametags. Each webbing assembly shall be identified with a flexible nametag sewn onto the webbing near the anchor fitting. The nametags shall contain the following information as a minimum.

a. Belt, Aircraft, HBU-12/A, P/N 549100.

b. Part Number (including the applicable revision letter) of the webbing assembly.

c. Month and year of manufacture (abbreviated as in 3.7.1.c, if desired).

3.8 Workmanship. All parts and assemblies of the lap belt shall be fabricated in a thoroughly workmanlike manner and shall be free of all defects that might adversely affect operation or strength. Appearance shall also be representative of best quality, and particular attention shall be given to freedom from burrs and sharp edges, heat treatment to specified strength, uniformity and adhesion of plating and solid film lubricants where specified, prevention of hydrogen embrittlement, proper alignment and secure retention of parts in assemblies. Webbing assemblies shall be free from loose, broken, or incomplete stitching, frayed or cut edges, and scuffed or melted areas other than hot-cut ends.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance shall not authorize submission of known defective material, either indicated or actual, nor shall it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (4.4).

b. First article inspection (4.5).

c. Quality conformance inspection (4.6).

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4.3 Inspection conditions

4.3.1 Test equipment and inspection facilities. The contractor shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.3.2 Atmospheric conditions. Unless otherwise specified, all measurements and tests shall be made at the ambient conditions of temperature, relative humidity and barometric pressure prevailing at the test site.

4.4 Materials inspection

4.4.1 Materials, processes, and parts. Materials, processes, and parts shall be inspected to ensure compliance with 3.3 and with the applicable requirements of the documents referenced herein.

4.4.1.1 Individual inspection of parts. Each ferrous casting shall be subjected to the radiographic examination described in 4.7.1.1. Each non-ferrous casting and load-carrying part other than pins and similar small turned parts with diameters of less than 0.25 inch shall be subjected after heat treatment, but before plating or coating to the magnetic particle or the dye penetrant examination described in 4.7.1.2, using whichever process is appropriate for the material involved.

4.5 First article inspection

4.5.1 First article test samples. First article test samples shall consist of at least three complete belts. The first articles shall be representative of the materials, processes, components, design, construction, and workmanship to be used in the production items. Each first article shall be identified with a tag that is plainly marked with the following information:

> Samples submitted by (contractor's name) on (date) for first article inspection in accordance with the requirements of (this specification number and applicable revision letter) under Contract No. _____.

4.5.2 First article testing. Unless otherwise specified in the contract or purchase order (see 6.2), the first articles shall be subject to all the examinations and tests specified in table I. The first article sample for group D inspection shall be subjected to the tests in the following sequence: 4.7.16, 4.7.13, 4.7.11, and 4.7.10. The other first article samples shall be subjected in any sequence desired by the contractor to all the group A, B, C, E and F inspections, except that within group E tests except 4.7.20 shall be conducted last.

4.5.3 First article inspection report. Unless otherwise specified in the contract (see 6.2), a first article inspection report shall be prepared in accordance with MIL-STD-831.

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4.6 Quality conformance inspection. Quality conformance inspections shall consist of the individual tests and sampling plan specified herein.

4.6.1 Inspection lot. A lot shall consist of the parts and assemblies for not more than 350 lap belts plus the belts which shall be subjected to the destructive lot acceptance tests (groups D, E and F) and thus become unfit for further use (see 4.7.21). All the belts in a lot shall have been produced for the contract under essentially the same conditions and submitted for acceptance at substantially the same time.

4.6.2 Individual tests. Each sample from each lot shall be subjected to the groups A and C inspections, as specified in table I.

4.6.2.1 Dimensions of parts inspection. The inspection level for the examination of dimensions shall be S-1 of MIL-STD-105. The Acceptance Quality Level (AQL) shall be 0.4 defects per hundred units (dhu) for critical defects, 1.0 dhu for major defects, and 4.0 dhu for minor defects. Classification of defects shall be in accordance with table II.

4.6.3 Sampling plans and tests

4.6.3.1 Group B inspections. Two samples from each lot shall be subjected to the group B inspections, as specified in table I.

4.6.3.1.1 Heat treated parts. The samples shall be subjected (before plating) to the hardness test described in 4.7.2.4. The inspection level shall be S-1 of MIL-STD-105. The AQL shall be 1.0 defects per 100 dhu. The sample unit and the unit for the expression of the lot size shall be the load-carrying parts for one belt.

4.6.3.1.2 Surface treated parts. The samples shall be subjected to the examination of surface treated (plated, coated, or passivated) parts described in 4.7.2.5. The inspection level shall be S-1 of MIL-STD-105. The AQL shall be 2.5 dhu. The sample unit and the unit for the expression of the lot size for the examination of surface treated parts shall be the surface treated parts of one belt.

4.6.3.2 Group D inspection. One sample per lot shall be subjected to the group D inspections as specified in table I. The acceptable quality levels for the group D tests shall be 1.0 dhu for the webbing creep test, combined proof load, and temperature tests, and zero dhu for the ultimate pressure and strength tests.

4.6.3.3 Group E inspection. One sample per lot shall be subjected to the group E inspections as specified in table I. The acceptable quality level for the group E tests shall be zero dhu for the 40-g dynamic load test with a negative-g strap, and one dhu for all other tests within group E.

4.6.3.4 Group F inspection. One sample per lot shall be subjected to the group F inspection listed in table I. The acceptable quality level for the group F test shall be zero dhu.

4.6.3.5 Disposition of belts subjected to destructive test. Belts which have been subjected to the group D, E and F inspections (see table I), shall be removed from service and tagged to indicate that they are unfit for further use.

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TABLE I. First article inspection.

First article	Applicable to	Requirement	Examination or Test
SAMPLE 1			
Group A inspections:			
Dimensions	Complete lap belt	3.4	4.7.2.3
Visual	Complete lap belt	3.8	4.7.2.1
Identification marking	Complete lap belt	3.7	4.7.2.2
Group B inspections:			
Hardness	Buckle hardware	3.4 & 3.8	4.7.2.4
Surface treatment	Buckle hardware	3.4 & 3.8	4.7.2.5
Group C inspections:			
No-load manual opening force	Buckle hardware	3.5.1	4.7.3
Static-load manual opening force	Buckle hardware	3.5.2	4.7.4
Static-load automatic disconnection pressure	Buckle hardware	3.5.3	4.7.5
Automatic disconnect leakage	Link Assembly	3.5.4	4.7.5.1
Gold key proof load	Buckle hardware	3.5.11	4.7.12
Webbing length adjustment force	Webbing assembly	3.5.13	4.7.14
Webbing assembly proof load	Webbing assembly	3.5.14	4.7.15
Group D inspections:			
Automatic disconnect ultimate pressure retention strength	Link assembly	3.5.10	4.7.11
Gold key ultimate strength	Buckle hardware	3.5.12	4.7.13
Webbing creep	Webbing assembly	3.5.15	4.7.16
Ultimate tensile strength	Buckle hardware	3.5.9	4.7.10

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TABLE I. First article inspection. - Continued

First article	Applicable to	Requirement	Examination or Test
SAMPLE 2			
Group A inspections:			
Dimensions	Complete lap belt	3.4	4.7.2.3
Visual	Complete lap belt	3.8	4.7.2.1
Identification marking	Complete lap belt	3.7	4.7.2.2
Group B inspections:			
Hardness	Buckle hardware	3.4 & 3.8	4.7.2.4
Surface treatment	Buckle hardware	3.4 & 3.8	4.7.2.5
Group C inspections:			
No-load manual opening force	Buckle hardware	3.5.1	4.7.3
Static-load manual opening force	Buckle hardware	3.5.2	4.7.4
Static-load automatic disconnection pressure	Buckle hardware	3.5.3	4.7.5
Automatic disconnect leakage	Link Assembly	3.5.4	4.7.5.1
Gold key proof load	Buckle hardware	3.5.11	4.7.12
Webbing length adjustment force	Webbing assembly	3.5.13	4.7.14
Webbing assembly proof load	Webbing assembly	3.5.14	4.7.15
Group E inspections:			
Manual release cycling	Buckle hardware	3.5.5	4.7.6
Vertical drop	Buckle hardware	3.5.6	4.7.7
Vibration	Buckle hardware	3.5.7	4.7.8
Shock/impact	Buckle harware	3.5.8	4.7.9
Combined proof loads	Complete lap belt	3.5.16	4.7.17
Combined proof loads without negative-g strap	Complete lap belt	3.5.16.1	4.7.17.1
High temperature	Complete lap belt	3.5.18.1	4.7.18.1
Low temperature	Complete lap belt	3.5.18.1	4.7.18.2
Humidity	Complete lap belt	3.5.18.2	4.7.19
Salt fog	Complete lap belt	3.5.18.2	4.7.19.1
Dust	Complete lap belt	3.5.18.2	4.7.19.2
40-g dynamic load with negative strap	Complete lap belt	3.5.17	4.7.20

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TABLE I. First article inspection. - Continued

First article	Applicable to	Requirement	Examination or Test
SAMPLE 3			
Group A inspections:			
Dimensions	Complete lap belt	3.4	4.7.2.3
Visual	Complete lap belt	3.8	4.7.2.1
Identification marking	Complete lap belt	3.7	4.7.2.2
Group B inspections:			
Hardness	Buckle hardware	3.4 & 3.8	4.7.2.4
Surface treatment	Buckle hardware	3.4 & 3.8	4.7.2.5
Group C inspections:			
No-load manual opening force	Buckle hardware	3.5.1	4.7.3
Static-load manual opening force	Buckle hardware	3.5.2	4.7.4
Static-load automatic disconnection pressure	Buckle hardware	3.5.3	4.7.5
Automatic disconnect leakage	Link Assembly	3.5.4	4.7.5.1
Gold key proof load	Buckle hardware	3.5.11	4.7.12
Webbing length adjustment force	Webbing assembly	3.5.13	4.7.14
Webbing assembly proof load	Webbing assembly	3.5.14	4.7.14
Group F inspections:			
Manual release cycling	Buckle hardware	3.5.5	4.7.6
Vertical drop	Buckle hardware	3.5.6	4.7.7
Vibration	Buckle hardware	3.5.7	4.7.8
Shock/impact	Buckle hardware	3.5.8	4.7.9
Combined proof loads	Complete lap belt	3.5.16	4.7.17
Combined proof loads without negative-g strap	Complete lap belt	3.5.16.1	4.7.17.1
High temperature	Complete lap belt	3.5.18.1	4.7.18.1
Low temperature	Complete lap belt	3.5.18.1	4.7.18.2
Humidity	Complete lap belt	3.5.18.2	4.7.19
Salt fog	Complete lap belt	3.5.18.2	4.7.19.1
Dust	Complete lap belt	3.5.18.2	4.7.19.2
40-g dynamic load without negative-g strap	Complete lap belt	3.5.17	4.7.20

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TABLE II. Classification of defects for dimensions.

Defects	Critical	Major	Minor
Any dimension with tolerance range of less than 0.010 inch not as specified (unless otherwise classified herein).	X		
Any dimension with tolerance range of more than 0.020 inch not as specified (unless otherwise classified herein).			X
Any thread related dimension (length and diameter of external or internal threads) not as specified.		X	
Any form or geometric tolerance not as specified.		X	
Any angle with tolerance range of 1° or less not as specified.		X	
Any angle with tolerance range of more than 1° not as specified.			X
Any radius or chamber dimension not as specified.			X
Any surface finish not as specified.			X
Any fractional dimension not as specified.			X

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4.7 Inspection methods

4.7.1 Materials tests

4.7.1.1 Radiographic examination. The load-carrying castings shall be x-rayed and examined in accordance with the class and grade of MIL-C-6021, that are specified on the drawings referenced herein. The interpretation of the x-rays shall be in accordance with ASTM E192.

4.7.1.2 Magnetic particle or dye penetrant examination. The load-carrying parts (after heat treatment but before plating or coating) shall be subjected to a magnetic particle examination in accordance with MIL-I-6868 or to a dye penetrant examination in accordance with MIL-I-6866, whichever is more appropriate to the material involved. There shall be no evidence of cracks or any other structural defects that could affect the operation or the strength of the hardware.

4.7.2 Examinations and tests

4.7.2.1 Visual examination. The belts shall be subjected to a visual examination (including the use of optical instruments if appropriate) to determine compliance with the requirements of this specification with respect to workmanship. The hardware shall be examined in particular to detect defects such as excessive pits, porosity, blistering, flaking, or any other evidence of inadequate adherence of the plating (and of solid film lubricant coating, if any is used), improper fit of parts, incompleteness (any part missing or improperly installed and secured), and misorientation (any part upside down or backwards). The webbing shall be examined in particular for defects such as loose, broken, or missing stitching, frayed or cut edges, and scuffed or melted areas other than hot-cut ends.

4.7.2.2 Examination of identification marking. The belts shall be examined to determine compliance with the identification marking specified in 3.7 thru 3.7.3. Particular attention shall be given to completeness, correctness, and legibility of the information included in the marking and to the location of marking.

4.7.2.3 Examination of dimensions. The parts shall be examined to determine conformance to the dimensions specified on the drawings referenced herein. Each dimension that is not as specified shall be classified as a defect in accordance with table II.

4.7.2.4 Hardness. Before being plated the load-carrying parts of the belt shall be subjected to a hardness test in accordance with ASTM E18, in order to determine compliance with the hardness requirements specified on the drawings referenced herein.

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4.7.2.5 Examination of surface treated parts. The finish of surface treated parts shall be visually examined for defects such as surface roughness, sharp edges, burrs, pitting, finish omitted or not as specified, evidence of corrosion, evidence of poor adhesion, supplementary treatment omitted or not as specified, inclusions that will produce voids in the finish if removed, excessive or uneven thickness of coating, scratches penetrating into base metal, and any other poor workmanship.

4.7.3 No-load manual opening force test. A tensile load of not more than 10 pounds shall be applied to the coupled release and link assemblies. The latch handle shall be blocked or wedged to hold it fully depressed relative to the hook. The opening force shall be applied under the finger-grip portion of the hook handle to lift it, using a spring scale which has maximum range of 0 to 25 pounds. The maximum force required to lift the hook handle and effect disconnection shall be not less than 2 pounds nor more than 12 pounds.

4.7.4 Static-load manual opening force test. This test shall be conducted in the same manner as specified in 4.7.2 except that the tension load applied to the coupled buckle hardware shall be 300 pounds and the spring scale used to measure the opening force shall have a range of 0 to 50 pounds maximum. The force required to effect disconnection shall be not less than 2 nor more than 30 pounds.

4.7.5 Static-load automatic disconnetion pressure test. A tensile load of 1000 pounds shall be applied to the coupled release and link assemblies. The automatic disconnect link assembly shall be conducted to a high pressure source or dry nitrogen with calibrated pressure gauge installed. Pressure of 75 psig shall be applied and shall not cause release. Pressure shall then be gradually increased until release is effected, which shall occur at not more than 750 psig.

4.7.5.1 Automatic disconnect leakage test. The disconnected link assembly shall then be immersed in soapy water and pressure increased to 2000 psig and held for 30 seconds. There shall be no observable leakage.

4.7.6 Manual release cycling test. The test article buckle hardware shall be subjected to 2500 cycles of manual release operation in which each cycle shall consist of engaging the manual release assembly with the automatic disconnect link assembly, applying to them a tensile load of 50 pounds, and sustaining this load while operating the manual release to disconnect them. After every 500 cycles this test article shall also be subjected to and pass the no-load manual opening force test (4.7.3).

4.7.7 Vertical drop test. A coupled release and link assembly shall be dropped from a height of 8 feet onto a concrete slab. This test shall be performed six times, changing the orientation of the test article by 90° each time to obtain impact in both directions along the horizontal, vertical, and transverse axes of the buckle hardware. No parts shall work loose and neither the manual release nor the automatic disconnect link assemblies shall be adversely affected enough to cause their disconnection. After the final drop the test article shall be subjected to and pass the no-load manual opening force test (4.7.3) and the static-load automatic disconnection pressure test (4.7.5).

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4.7.8 Vibration test. A coupled release and link assembly shall be installed with approximately 10-pound tension loading in a fixture which shall then be mounted on a suitable vibration tester as required to subject the buckle hardware successively to vibration along its horizontal, vertical, and transverse axes. In each position, the hardware shall be subjected for 30 minutes to linear vertical vibration which shall be varied continuously from 10 to 55 Hz at approximately 1/2 octave per minute, with total table excursion of approximately 0.03 inch. No parts shall work loose and neither the manual release nor the automatic disconnect link assemblies shall be adversely affected enough to cause their disconnection. After the final test the test article shall be subjected to and pass the no-load manual opening force test (4.7.3) and the static-load automatic disconnection pressure test (4.7.5).

4.7.9 Shock/impact test. The same test article and fixture shall then be used to subject the hardware to shock/impact in both directions along the same three axes, applying in each of the six tests a pulse with approximately 3-millisecond base and 400g ±40-g peak. Video-taped motion pictures (500 frames per second) shall be taken to show the respective initial impacts. No parts shall work loose and neither the manual release nor the automatic disconnect link assemblies shall be adversely affected enough to cause their disconnection. After the final impact the test article shall be subjected to and pass the no-load manual opening force test (4.7.3) and the static-load automatic disconnection pressure test (4.7.5).

4.7.10 Buckle hardware ultimate strength tensile test. Suitable lengths of webbing having thickness of less than 1/8 inch and breaking strength of at least 7000 pounds shall be looped around the webbing crossbars of the coupled release and link assemblies in a manner simulating attachment of the webbing assemblies to them. Both ends of these webbing loops at each end of the buckle hardware shall be gripped by the corresponding jaws of the testing machine. Tensile load shall then be applied to the test article at jaw separation rate of approximately 4 inches per minute until the load has reached 7000 pounds. After being held at this value for 30 seconds, the load shall be reduced to not more than 10 pounds and the no-load opening force test (4.7.3) repeated. The force required to effect release shall be not less than 2 nor more than 12 pounds. No structural failure shall result from this ultimate strength test, but permanent deformation is permissible.

4.7.11 Automatic disconnect ultimate pressure retention strength test. Using water or oil as the working fluid, hydrostatic pressure of 7500 psig shall be applied to the automatic disconnect portion of the link assembly and held for 5 seconds. There shall be no structural failure and leakage shall not exceed 5 drops of fluid, but permanent deformation is permissible.

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4.7.12 Gold key proof load test. The gold key shall be installed on a link assembly tongue that is then coupled to a manual release assembly. The latter shall be mounted by its crossbar holes in the test apparatus in a manner such that the proof load will be applied in the plane of symmetry and the assembly can pivot freely in response to this loading. Removal force of 75 pounds sustained for at least 5 seconds shall be applied to the key via its lanyard or an equivalent tie directed away from the manual release. There shall be no effect on the key, its lanyard (if one is installed), or the buckle hardware components.

4.7.13 Gold key ultimate strength test. The test specified in 4.7.12 shall be repeated, except that the removal force applied to the key shall be 450 pounds and be sustained for 30 seconds. Permanent deformation is permissible, but the key and tongue shall remain engaged with the manual release assembly and there shall be no structural failure in the components involved.

4.7.14 Webbing length adjustment force test. A 25-pound load shall be applied to the webbing assembly and a spring scale with range of 0 to 50 pounds maximum shall be connected to the free end as shown in figure 1. The free end of the webbing shall then be pulled downward by means of the spring scale until the weight has been lifted free of its support. The reading on the belt tension spring scale as this is accomplished shall not exceed 50 pounds. The spring scale shall then be attached to the release tab of the adjuster and pulled downward to release the suspended load which shall require not more than 15 pounds force as read on the spring scale.

4.7.15 Webbing assembly proof load test. The webbing assembly shall be arranged for test as shown in figure 1 except that it shall be mounted in tensile test apparatus capable of applying 1000-pound load to the test article. The distance between the doubled-over portion of the free end of the webbing and the adjuster shall be at least 2 inches and shall be measured before and after load of 1000 pounds has been applied to the webbing assembly and sustained for 30 seconds. This proof loading shall have no adverse effect upon any element of the webbing assembly and webbing slippage through the adjuster shall not exceed 1/2 inch.

4.7.16 Webbing creep test. A complete lap belt (or its coupled webbing assemblies) shall be arranged for tensile loading over a block simulating the cross section of a torso, as shown in figure 2. The outboard end of one webbing assembly shall be attached to a fixed support. The outboard end of the other webbing assembly shall be connected to a load-application device with cycling stroke such that belt tension load of 226 pounds will be applied and reduced to zero at the rate of 60 ±10 cycles per minute, with the webbing to be completely slack during the zero load portion of each cycle. Each webbing shall be pulled through its adjuster to eliminate slack and any set in the webbing before marking it for measurement of creep and beginning the test. The belt shall then be subjected to 2500 complete test cycles with this loading. Webbing creep through either adjuster shall not exceed 1/2 inch.

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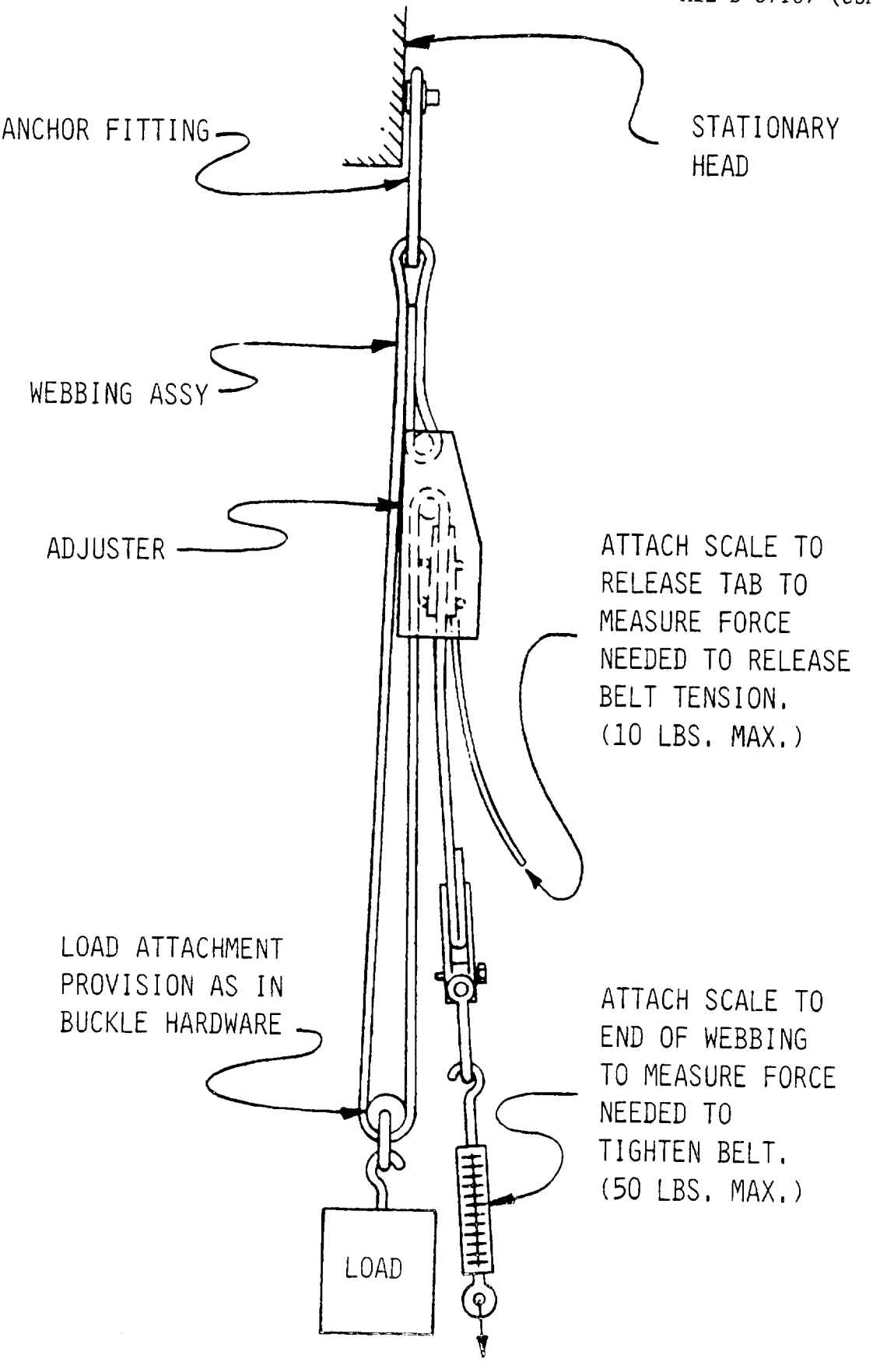
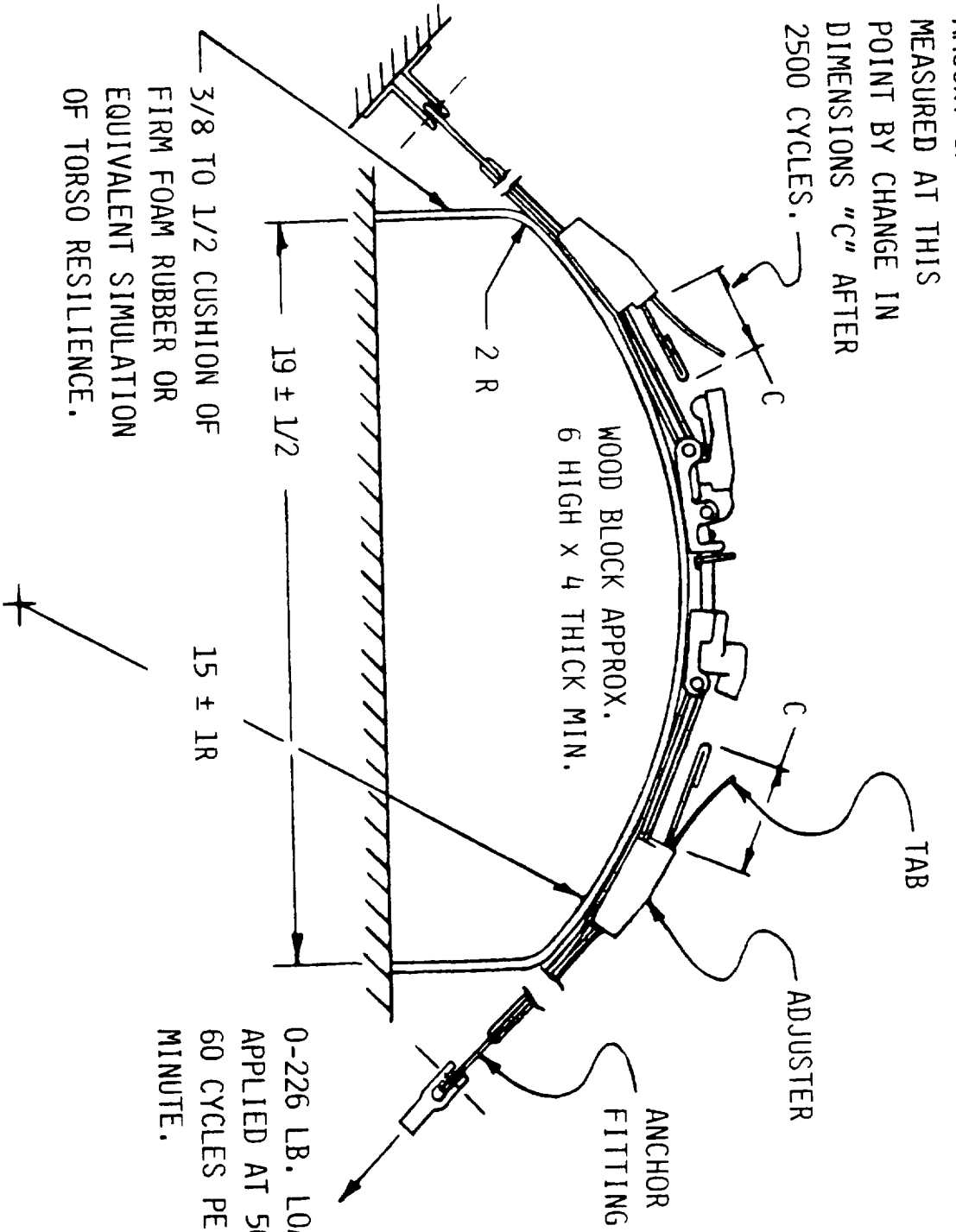


FIGURE 1. Setup for adjustment force tests.

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NOTE: Dimensions are in inches.

FIGURE 2. Webbing adjuster creep test setup.

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4.7.17 Lap belt combined proof load test. A complete lap belt assembly shall be arranged for this test by installing (in this sequence) a negative-g strap, a pair of oppositely-directed shoulder harness straps, and a gold key (if one is required) on the tongue of its link assembly before coupling the latter to the manual release assembly. Substitute methods of applying the restraint strap loads to the buckle hardware may be used provided that they simulate the configuration of the straps' end loops. The lap belt shall be arranged over a block simulating the cross section of a torso, such as that used in the 4.7.16 test. Tensile load shall be applied to the belt via an anchor fitting (or the fittings) at a jaw separation rate of approximately 4 inches per minute until the load reaches 2880 pounds. The negative-g strap load and the counter-balancing loads in the two shoulder harness straps shall be applied in the plane of the link tongue, but perpendicular to it, to produce and maintain 3600 pound transverse load. These combined loads shall be sustained for 30 seconds without having adverse effect on any component of the belt nor resulting in more than 1/2-inch slippage of the webbing through either adjuster. After this proof loading, the belt shall be subjected to and pass the 4.7.4 and 4.7.5 static-load manual release and automatic disconnection pressure tests.

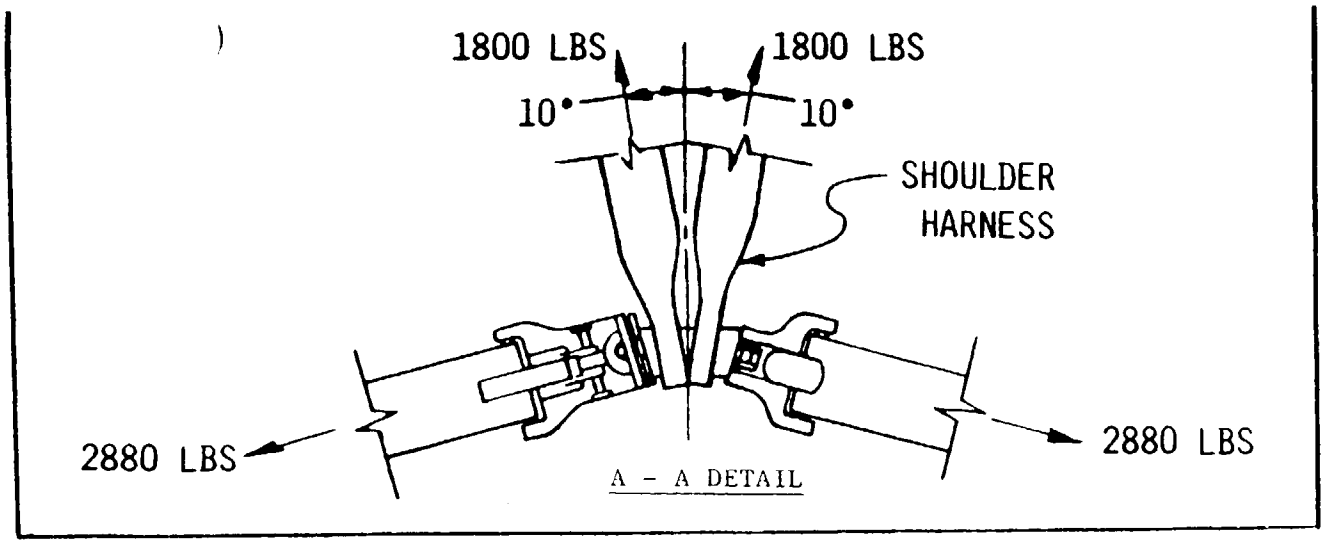
4.7.17.1 Combined proof load test without negative-g strap. A lap belt shall be tested as specified in the 4.7.17 test, except that the negative-g strap and its loading shall be omitted. The torso block shall also be configured and the test setup arranged as shown on figure 3, so that the angular relationship between the simultaneously applied belt and shoulder-harness strap loads can change as they are increased to their respective maximum values, and thus be reasonably representative of their relationship during the deceleration condition described in 3.5.17.

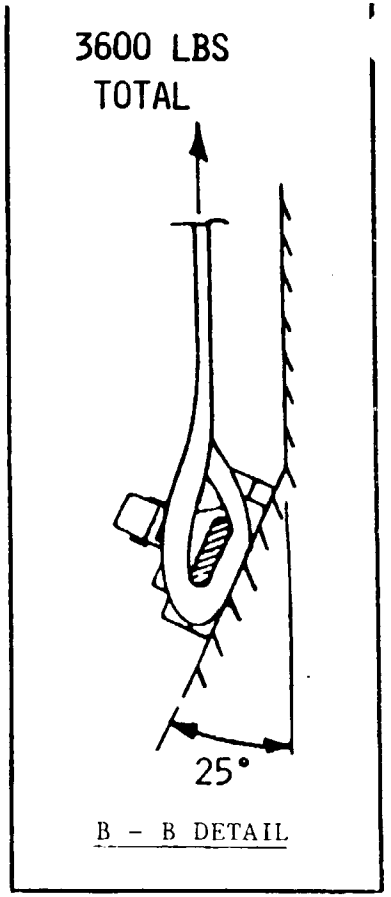
4.7.18 Temperature tests

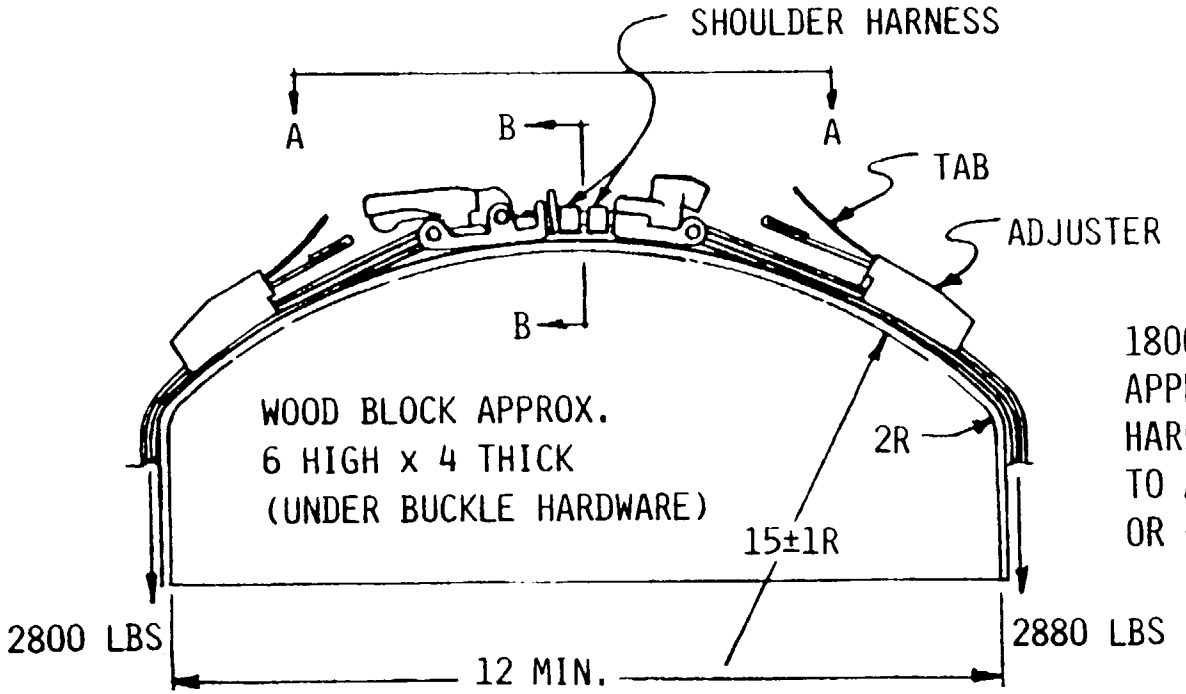
4.7.18.1 High temperature test. Unless otherwise specified in the contract or purchase order (see 6.2), a complete belt assembly shall be conditioned at a temperature of 200°F for at least 3 hours in accordance with procedure II, method 501.2 of MIL-STD-810. As quickly as possible after its removal from the temperature chamber, the belt shall be mounted in a tension tester and its automatic disconnect link assembly connected to a source of high-pressure dry nitrogen gas via a suitable hose and a pressure regulator with calibrated pressure gauge. The belt halves shall then be quickly disconnected and reconnected to check operation of the manual release, and belt slack in the setup shall be removed by pulling the webbing through both adjusters to check their operation. Tensile load of 1800 pounds shall then be applied to the belt and the gas pressure increased until disconnection occurs. This test shall be completed within 4 minutes after the belt has been removed from the temperature chamber. Webbing slippage shall not exceed 1/2 inch through either adjuster and disconnection shall require pressure of not less than 75 psi nor more than 750 psi.

4.7.18.2 Low temperature test. The test specified in 4.7.18.1 shall be repeated with the same test article, except that the belt shall have been conditioned at a temperature of -65°F to -70°F for at least 3 hours in accordance with procedure II, method 502.2 of MIL-STD-810. Compliance with the post-conditioning requirements of 4.7.18.1 shall be demonstrated.

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1800 LBS TO BE APPLIED BY SHOULDER HARNESS STRAPS CONFORMING TO AF DRAWING 56D533, OR EQUIVALENT.

NOTE: Dimensions are in inches.

FIGURE 3. Setup for combined-load proof test without negative-g strap.

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4.7.19 Humidity test. Unless otherwise specified in the contract or purchase order (see 6.2), a first article belt assembly shall be subjected for 48 hours to humidity conditioning in accordance with procedure I, method 507.1 of MIL-STD-810. Before and after this conditioning, and while installed on the belt, the buckle hardware and both adjusters shall be subjected to and pass the tests described in 4.7.4, 4.7.5, and 4.7.14.

4.7.20 Salt fog test. The test specified in 4.7.19 shall be repeated with the same test article except that the belt shall have been subjected for 48 hours to salt fog conditioning in accordance with procedure I, method 509.1 of MIL-STD-810. Compliance with the post-conditioning requirements of 4.7.19 shall be demonstrated.

4.7.21 Dust test. The test specified in 4.7.19 shall be repeated with the same test article except that the belt shall have been subjected for 28 hours to dust conditioning in accordance with procedure I, method 510.1 of MIL-STD-810. Compliance with the post-conditioning requirements of 4.7.19 shall be demonstrated.

4.7.22 40-g dynamic load test. Unless otherwise specified in the contract or purchase order (see 6.2), the belt assemblies shall be subjected to two 40-g dynamic load tests, conducted as follows.

4.7.22.1 In both inspections, the acceleration shall be applied to a seated 95th percentile size anthropomorphic dummy restrained by the lap belt, and two USAF shoulder straps. One test shall be conducted with a USAF negative-g strap and one test shall be conducted without a negative-g strap. The restraint system shall be mounted on a rigid test seat representative of USAF ejection seat size and geometry. The restraint system shall be pre-tensioned to 10 ±2 pounds, measured at each restraint strap tiedown point.

4.7.22.2 Acceleration and instrumentation requirements. The acceleration-time profile shall be triangular shaped with a velocity change of at least 64 feet per second and a time-to-peak acceleration of not more than 0.07 seconds. The acceleration, which shall be applied in the X-axis (chest-to-back), shall be measured using a translational accelerometer with a frequency response that is flat, within ±5 percent, from DC to 100 Hz. Data transmission and recording equipment shall have a frequency response which is flat, within ±5 percent, and in excess of 2.5 times the roll-off frequency of the accelerometer.

4.7.22.3 Acceptance criteria. The acceleration test shall not cause any structural failures of the lap belt assembly. Webbing slippage that does not exceed 1/2 inch per adjuster and other such non-catastrophic reactions of lap belt components that would not cause injury to the restrained occupant are permissible. The lap belt shall be operable to permit manual release of the restraint system immediately after the test. This shall require manual opening force of not less than 2 pounds nor more than 12 pounds when performed as described in 4.7.3.

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4.8 Noncompliance. If a sample fails to pass group D, group E or group F inspections (see table I), the manufacturer shall notify the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of the product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the inspection activity has been taken. After the corrective action has been taken, group D, E, or F inspections shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A, B, and C inspections and tests may be reinstituted; however, final acceptance and shipment shall be withheld until the group D, E or F inspections has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.9 Examination of packaging. Shipping containers fully prepared for delivery except for closure shall be examined for defects listed in table III in order to determine compliance with the packaging requirements. The sample unit and the unit for the expression of the lot size for the examination of packaging shall be one shipping container fully prepared for delivery except for closure. The inspection level for the examination of packaging shall be S-3 of MIL-STD-105. The AQL for the examination of packaging shall be 2.5 dhu.

TABLE III. Examination of packaging.

Examine	Defects
Marking (exterior and interior)	Marking omitted, incorrect, incomplete, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or not as specified. Any component damaged or otherwise defective.
Workmanship	Inadequate application of components, such as incomplete closure of case liners or container flaps, improper taping, loose strapping, or inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross or net weight not as specified.
Contents	Quantity of belts per container more or less than specified.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each belt shall be preserved in accordance with Method III of MIL-P-116.

5.1.2 Commercial. The commercial preservation of belts shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or Commercial, as specified (see 6.2).

5.2.1 Level A. The packaged belts shall be packed in containers conforming to PPP-B-601 overseas type, or PPP-B-621, overseas type. Closure and strapping shall be in accordance with applicable container specifications.

5.2.2 Level B. Level B packing shall be the same as Level a packing except that the shipping container shall be class domestic, or a container shall be used that conforms to class weather-resistant of PPP-B-636.

5.2.3 Level C. The packaged belts shall be packed in fiberboard containers conforming to PPP-B-636, class domestic.

5.2.4 Commercial. The preserved belts shall be packed in accordance with the requirements of ASTM D 3951.

5.4 Marking

5.3.1 Levels A, B and C. In addition to any special or other identification marking requirements by the contract or purchase order (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The lap belts covered by this specification are intended for use in aircraft ejection seats in which a back- or seat-pack parachute is used by the occupant and the overall restraint system may include negative-g strap(s) in addition to the shoulder harness, with the disconnectable ends of all such straps to be engaged with the belt buckle hardware when it is joined. In normal use, the belt halves and these other restraint straps are coupled and uncoupled manually, but are to disconnect automatically after seat ejection when the buckle hardware receives hot-gas pressure from a delayed firing initiator that is triggered during the ejection sequence. In some of these ejection systems the subsequent separation of the seat from its occupant initiates automatic opening of his parachute, because the ripcord is connected to a gold key that is locked to the lap belt when it is opened automatically.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article requirements (see 3.1, 4.5.1, 4.5.3, 4.7.18, 4.7.19, and 4.7.20).

c. National stock numbers, serial numbers, and specification number if any of these are to be shown on the nameplate (see 3.7.1).

d. Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 First articles. First articles shall be inspected and approved under the appropriate provisions of 7-104.55 of the Department of Defense Procurement Regulations. The first articles should be at least three preproduction samples and, if required, additional samples to be subjected to the group D, E and F inspections (see 4.5 and 4.5.1). In all acquisition documents, the contracting officer should include specific instructions regarding arrangements for inspection, approval, and disposition of the first articles, and of the lot samples which are subjected to the group D inspections.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82, 99

Preparing activity:
Air Force - 11

Project No. 1680-0518

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(See Instructions – Reverse Side)

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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